Week 1 Progress Report

From Chaos to Prediction: Machine Learning and Physics-Informed Neural Networks

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Scope of this report. This document summarizes the tasks completed during Week 1 of the internship: (1) simulation and characterization of discrete and continuous chaotic systems, (2) quantitative confirmation of chaos in the Lorenz attractor via five independent diagnostics, (3) a mini literature review of eight published papers on AI applied to chaotic/time-series prediction, (4) an ECG signal preprocessing and R-peak detection pipeline on the MIT-BIH Arrhythmia Database, and (5) an extended literature search on ECG evaluation protocols and hybrid deep learning architectures.

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1 Overview

During the first week of the internship, work was organized around building the foundational skills needed for the track *"From Chaos to Prediction: Machine Learning and Physics-Informed Neural Networks."* Two parallel tracks were pursued: a theoretical/numerical track on chaotic dynamical systems (Tasks 1–3), and an applied biomedical signal-processing track on ECG data from the MIT-BIH Arrhythmia Database (Tasks 4–5 / `nu-lab1`), which will feed into later reservoir-computing and physics-informed modeling stages of the project.

Task	Title	Deliverable
Task 1	Discrete & continuous chaotic systems	Executed Jupyter notebook
Task 2	Quantitative chaos diagnostics (Lorenz)	Notebook + formal report
Task 3	Mini literature review (8 papers)	Structured spreadsheet
Task 4	ECG preprocessing & R-peak detection	Kaggle notebook (`nu-lab1`)
Task 5	Data-splitting protocols & extended literature search	Updated spreadsheet + architecture notes

2 Task 1: Discrete and Continuous Chaotic Systems

The goal of Task 1 was to build intuition for chaotic behavior by simulating representative discrete maps and continuous flows, and to study how the systems' sensitivity to parameters and numerical step size manifests in practice. The work is organized into three parts.

2.1 Part A – Discrete Chaotic Systems

- **Logistic map** $x_{n+1} = r\,x_n(1 - x_n)$ was iterated with $x_0 = 0.5$, $r = 3.9$, for 1000 iterations (200 discarded as transient). The resulting time series never settles to a fixed point or periodic cycle, which is the classic signature of deterministic chaos at $r = 3.9$.
- **Hénon map** $x_{n+1} = 1 - ax_n^2 + y_n,\ y_{n+1} = bx_n$ was simulated with $a = 1.4$, $b = 0.3$ over 5000 iterations. The attractor traces a bounded, self-similar, fractal-like set rather than filling the plane or collapsing to a curve.
- **Bifurcation diagrams** were produced for both maps by sweeping the control parameter ($r \in [2.5, 4]$ for the logistic map, step 0.001). The diagrams reveal the expected period-doubling cascade into chaos beyond $r \approx 3.57$, including narrow periodic windows (e.g. the period-3 window near $r \approx 3.83$).

2.2 Part B – Continuous Chaotic Systems

Four continuous systems were integrated (RK4) and visualized in 3D phase space: **Lorenz**, **Rössler**, **Chen**, and the **4-D hyperchaotic Rössler** system. Each system's characteristic attractor geometry (e.g. the Lorenz butterfly, the Rössler single-scroll, the Chen double-scroll) was reproduced and discussed.

2.3 Part C – Sensitivity Analysis

- **Parameter sweep:** the Lorenz ρ parameter was swept and the local maxima of $z(t)$ were tracked, showing how the qualitative structure of the attractor changes with ρ.
- **Step-size sensitivity:** the same Lorenz trajectory was re-integrated at several RK4 step sizes dt, using the smallest dt as the accuracy baseline, to quantify how numerical step size alone can act as a source of divergence in chaotic simulations.

2.4 Key Takeaways

- Discrete maps have no intrinsic notion of "time between steps"; chaos is read off directly from the iterates and bifurcation structure. Continuous systems introduce an additional numerical-sensitivity axis (step size) that does not exist for discrete maps.
- A standard chaotic system (Lorenz, Rössler, Chen) has exactly one positive Lyapunov exponent, while a hyperchaotic system (4-D Rössler) has two or more, producing richer and harder-to-predict dynamics.

3 Task 2: Quantitative Chaos Diagnostics on the Lorenz Attractor

Building on Task 1's qualitative (visual) argument, Task 2 quantitatively confirms chaos in the Lorenz system ($\sigma = 10$, $\rho = 28$, $\beta = 8/3$, initial condition $(1, 1, 1)$, RK4, $dt = 0.01$) using five independent numerical diagnostics. A long trajectory of $T = 3000$ time units (300,000 steps, first 20 discarded as transient) was generated to support the statistical estimators.

3.1 Method A – Phase Space Reconstruction

Using only the scalar observable $x(t)$, Takens' embedding theorem was applied to reconstruct delay vectors $\mathbf{y}(t) = [x(t), x(t + \tau), \ldots, x(t + (m - 1)\tau)]$.

- **Time delay (AMI):** first local minimum of the average mutual information curve gives $\tau_{\text{opt}} = 16$ samples – within the expected range for Lorenz ($\tau \approx 10$–17).
- **Embedding dimension (FNN):** the false-nearest-neighbour percentage collapses from 99.48% at $m = 1$ to 0.00% at $m = 3$, giving $m_{\text{opt}} = 3$ – exactly matching the true state-space dimension of the Lorenz system.

3.2 Method B – Lyapunov Exponent

Computed two independent ways:

- **From the ODE** (Wolf's Gram–Schmidt reorthogonalization on the analytic Jacobian): full spectrum $\lambda = (+0.9005,\ -0.0029,\ -14.5642)$ nats/time unit. The sum of exponents matches the trace of the Jacobian, $-(\sigma + 1 + \beta) = -13.6667$, to 4 significant figures – a strong internal consistency check.
- **From the scalar time series** (Wolf's phase-space tracking method, using $\tau = 16$, $m = 3$ from Method A): $\lambda_1 = 0.9523$ nats/time unit.
- The two estimates agree to within 5.8%, and both are clearly positive, confirming exponential sensitivity to initial conditions from two independent routes.

3.3 Method C – Kolmogorov (K2) Entropy

Estimated via the Grassberger–Procaccia relation from the correlation sum across embedding dimensions $m = 3$ to 10. K2 flattens to ≈ 1.83 nats/time unit (mean of $m = 8, 9, 10$) – finite and positive, consistent with deterministic chaos (a periodic system would give $K2 = 0$; pure noise would diverge as m grows).

3.4 Method D – Correlation Dimension

The correlation dimension D_2 (Grassberger–Procaccia slope of $\ln C(r)$ vs. $\ln r$) was computed for $m = 2$ to 10 and saturates at $D_2 \approx 2.12$ (mean of $m = 8, 9, 10$) – close to the textbook Lorenz value ($D_2 \approx 2.05$) and clearly non-integer, evidencing a genuine fractal attractor.

3.5 Method E – Box-Counting (Fractal) Dimension

Using a denser trajectory ($dt = 0.005$) on the raw 3D state, the box-counting dimension was found to be $D_0 = 1.843$. This is slightly below D_2, a known finite-sample artefact of box counting rather than evidence against the fractal structure already confirmed in Method D.

3.6 Summary Table

Metric	Result
Time delay τ (AMI)	16 samples
Embedding dimension m (FNN)	3
Largest Lyapunov exp. – ODE (Wolf GSR)	$\approx +0.90$ nats/time unit
Largest Lyapunov exp. – time series (Wolf MLE)	$\approx +0.95$ nats/time unit
K2 entropy	≈ 1.83 nats/time unit (finite, positive)
Correlation dimension D_2	≈ 2.10–2.15 (saturated, non-integer)
Box-counting dimension D_0	≈ 1.84 (finite-sample-limited)

Conclusion: all five independent methods agree – the Lorenz system at these parameters is chaotic, with exactly one positive Lyapunov exponent (chaotic, not hyperchaotic), a finite positive K2 entropy, and a saturating, non-integer fractal dimension. This replaces the purely visual chaos argument from Task 1 with rigorous quantitative evidence. A full standalone report was also written up for this task.

4 Task 3: Mini Literature Review – AI for Chaotic Time-Series Prediction

A structured literature review of eight formally published, peer-reviewed papers on applying AI/deep learning to chaotic systems and time-series forecasting was compiled into a spreadsheet, with each entry including the venue, year, journal quartile, impact factor / CiteScore, a summary, limitations, and a personal relevance comment.

4.1 Papers Reviewed

1. **TCN-Linear hybrid model** for chaotic time-series forecasting (*Entropy*, MDPI, 2024, Q2) – combines dilated causal convolutions (TCN) with an LSTF-Linear model, benchmarked against Bi-LSTM, Seq2Seq, CNN-GRU.
2. **LSTM vs. Transformer comparison** on chaotic orbits spanning a wide range of Lyapunov exponents (*Chaos, Solitons & Fractals*, 2025, Q1) – LSTNet gives the best overall results.
3. **Physics-informed neural operator (PINO)** for chaotic time series (*Chaos, Solitons & Fractals*, 2024, Q1) – hybrid data-physics training outperforms purely data-driven or physics-driven variants under noisy data.
4. **AI-Lorenz**, a physics-data-driven framework using symbolic regression to recover governing equations of chaotic systems (*Chaos, Solitons & Fractals*, 2024, Q1).
5. **Physics-informed hierarchical echo state network (Pi-HESN)** (*Expert Systems with Applications*, 2023, Q1) – stacked reservoirs with physics-informed loss, reducing required training data.
6. **P2P / NAR / S2S comparison** across MLP, LSTM, and Fourier Neural Operator backbones (*Nonlinear Dynamics*, Springer, 2026, Q1) – evaluates both short-term accuracy and long-term attractor fidelity.
7. **Large-scale ML benchmark** for low-dimensional chaotic systems (*Chaos: An Interdisciplinary Journal of Nonlinear Science*, 2026, Q2) – compares reservoir computing, LSTMs, and Transformer/N-BEATS models with a new "cumulative maximum error" metric.
8. **Spatial + temporal deep learning** combining raw time series with recurrence plots (*Nonlinear Dynamics*, Springer, 2025, Q1) – validated on Lorenz, Rössler, and real ECG (arrhythmia) data.

4.2 Key Insights for the Project

- Hybrid architectures (TCN-linear, physics-informed operators, physics-informed reservoir networks) consistently outperform single-paradigm models on chaotic series, motivating the project's physics-informed / reservoir computing direction.
- Forecasting performance for all reviewed methods degrades as the largest Lyapunov exponent increases – sensitivity to initial conditions is a fundamental limit, not just a data or architecture problem.
- Long-term phase-space (attractor) fidelity is emerging as a more chaos-theoretically meaningful evaluation criterion than short-term point-forecast error alone.

5 Task 4 (nu-lab1): ECG Preprocessing and R-peak Detection

This task builds a rigorous ECG preprocessing pipeline on the MIT-BIH Arrhythmia Database (48 records), aligned to a specific paper's filter-bank structure, as groundwork for later chaos/reservoir-computing analysis of biomedical signals.

5.1 Data Characterization

Each record's sampling frequency, signal length, lead configuration, and beat annotations were inspected. Beat-symbol counts were aggregated across all 48 records to understand the overall class distribution. A per-record noise profile (baseline-wander %, powerline %, high-frequency %) was computed via Welch PSD, identifying **Record 108** as the worst case for baseline wander (39.78% of signal power) and **Record 214** as the cleanest (0.49%).

5.2 Filtering Pipeline

Several denoising variants were implemented and compared on the noisiest record:

- Butterworth high-pass, low-pass, and band-pass (0.5–35 Hz) filters.
- A 60 Hz IIR notch filter, and an alternative Kaiser-window FIR band-stop filter for powerline interference removal.
- Wavelet-based denoising via both **Stationary Wavelet Transform (SWT)** and **Discrete Wavelet Transform (DWT)** (wavelet `rbio3.9`, level 5), following the paper-optimal configuration, with soft thresholding based on the median absolute deviation of the detail coefficients.

Each variant was evaluated on baseline-wander /powerline / high-frequency power fractions, SNR improvement, and correlation with the raw signal, across multiple records (108, 121, 111, 222).

5.3 R-peak Detection and Morphology

- R-peaks were detected on the SWT-cleaned signals using the WFDB `XQRS` detector.
- Detection was validated against the database's expert beat annotations (within a 100 ms tolerance), reporting sensitivity and precision per record.
- Beat morphology (P-wave, QRS, ST-segment, T-wave amplitudes, and R-amplitude) was extracted in fixed windows around each detected R-peak, and compared across raw, band-pass+notch, SWT, and DWT variants to quantify how much of each morphological feature is preserved (as a % of the raw signal) by each denoising method.

5.4 Status

The preprocessing pipeline, noise characterization, filter-bank comparison, R-peak detection, and morphology-preservation analysis are complete for the representative records tested; this pipeline feeds directly into Task 5's methodological search and will be extended to the full 48-record database for the reservoir-computing / physics-informed modeling stages planned next.

6 Task 5: Data-Splitting Protocols and Extended Literature Search

With a working preprocessing pipeline in place from Task 4, Task 5 turns to the methodological question of how the resulting ECG beats should be split into training and test sets, and broadens the literature search to identify hybrid deep-learning architectures that could be adapted for ECG modeling.

6.1 Patient-Wise vs. Signal-Wise Splitting

The distinction between **inter-patient (patient-wise)** and **intra-patient (signal-wise)** evaluation protocols for ECG datasets was investigated:

- In **intra-patient (signal-wise) splitting**, beats from the same patient record can appear in both the training and test sets. Because beat-to-beat morphology is highly correlated within a single patient, this protocol tends to *inflate* reported accuracy and does not reflect how a model would perform on an unseen patient.
- In **inter-patient (patient-wise) splitting**, all beats from a given patient are confined to either the training set or the test set, never both. This is a substantially harder and more realistic test of generalization, since the model must cope with inter-subject variability in ECG morphology it has never seen during training.
- The practical effect on evaluation was examined qualitatively: models that look strong under signal-wise splitting typically show a marked performance drop under patient-wise splitting, which is why patient-wise protocols are considered the more reliable and clinically meaningful benchmark.

6.2 Literature Review on Evaluation Protocols

A focused literature search was carried out on papers that explicitly discuss or adopt inter-/intra-patient evaluation protocols for arrhythmia classification, with attention to each paper's dataset, preprocessing pipeline, splitting strategy, model architecture, reported limitations, and suggested future work. Newly identified papers were added to the running literature-review spreadsheet (extending the Task 3 sheet) with these fields populated for each entry.

6.3 Architecture Study: HAR-former

As a first step toward exploring hybrid architectures for ECG analysis, the paper *HAR-former: Hybrid Transformer with an Adaptive Time-Frequency Representation Matrix for Long-Term Series Forecasting* was studied in detail. The goal was to extract its core design ideas – notably its adaptive time-frequency representation matrix and its hybrid Transformer structure for long-term series forecasting – and to assess whether analogous time-frequency-aware hybrid designs could be adapted to ECG beat classification or arrhythmia forecasting.

6.4 Brainstorming Novel Hybrid Architectures

Building on the expanded literature review and the HAR-former study, candidate directions for a novel hybrid deep-learning architecture for ECG analysis were brainstormed. Each candidate idea was cross-checked against the literature collected so far to confirm it had not already been explored in the reviewed papers, in order to identify a genuinely open research direction rather than duplicating existing work.

6.5 Status

The comparison of splitting protocols, the expanded literature-review spreadsheet, and the HAR-former architecture study are complete; the brainstormed hybrid-model directions will be narrowed down and validated against the ECG preprocessing pipeline from Task 4 in the coming days.

7 Summary and Next Steps

- **Task 1** established a working simulation toolkit for discrete and continuous chaotic systems and their sensitivity behavior.
- **Task 2** rigorously confirmed chaos in the Lorenz system using five independent, cross-validated quantitative diagnostics.
- **Task 3** surveyed the current state of AI-based chaotic time-series forecasting, directly motivating a physics-informed / hybrid modeling direction.
- **Task 4 (nu-lab1)** delivered a validated ECG preprocessing and R-peak detection pipeline on real MIT-BIH data, bridging the chaos-theory track with the biomedical-signal track of the project.
- **Task 5** clarified the patient-wise vs. signal-wise evaluation question, expanded the literature review, and produced an initial set of candidate novel hybrid architectures for ECG modeling, informed by the HAR-former study.

Planned next steps: extend the ECG pipeline to the full MIT-BIH database under a patient-wise splitting protocol, begin applying reservoir-computing / physics-informed models to both the Lorenz benchmark and the preprocessed ECG signals, narrow down and validate the brainstormed hybrid architectures from Task 5, and use the literature review's methodological insights (attractor-fidelity metrics, hybrid architectures, evaluation protocols) to guide model selection.